

First-principles perturbative theory of anomalous scaling in a stochastic shell model of turbulence

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Abstract

Deriving anomalous scaling exponents from the equations of motion remains a central problem in the statistical theory of turbulence. Here we obtain a first-principles perturbative solution for a nonlinear stochastic dyadic shell model. The model preserves the conservative cascade structure and exact scaling symmetry of the deterministic dynamics, while stochastic transfer fluctuations provide a perturbative setting in which the leading-order rescaled dynamics is Gaussian. Using the statistically restored hidden scaling symmetry of the inertial-range equations, we determine the stationary statistics of the rescaled variables. We then formulate anomalous scaling as a Perron–Frobenius eigenvalue problem for the multiplier statistics. The resulting perturbative expansion yields explicit analytical expressions for the scaling exponents of structure functions of arbitrary order in the weak-noise regime. Direct numerical simulations provide an independent verification of the theoretical predictions. The results demonstrate that the hidden-symmetry perturbation framework extends from linear random models to a genuinely nonlinear cascade system.

1 Introduction

Intermittency in developed turbulence is a prototypical example of anomalous scaling in a strongly nonequilibrium multiscale system. It is manifested by the nonlinear dependence of the scaling exponents of stationary structure functions on their order. For velocity increments over a scale ℓ , one writes $S_p(\ell) = \langle |\delta u(\ell)|^p \rangle \sim \ell^{\zeta_p}$. The Kolmogorov 1941 theory [20, 16] predicts the dimensional exponents $\zeta_p = p/3$, whereas experiments and numerical simulations reveal systematic deviations from this law. Deriving these anomalous exponents analytically

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from the governing dynamics remains one of the principal problems in the statistical theory of turbulence.

Kolmogorov’s refined similarity hypothesis [21] relates fluctuations of velocity increments to those of the locally averaged energy dissipation. In a discrete cascade formulation, this viewpoint leads naturally to random multipliers connecting neighboring scales [1, 7, 13]. The multifractal formalism of Parisi and Frisch [35, 16] represents the flow as a superposition of regions characterized by different local scaling exponents and expresses the structure-function exponents through the corresponding spectrum of singularities. These approaches provide successful phenomenological descriptions of intermittency [17, 11, 24, 2], but the multiplier statistics or the multifractal spectrum are not determined directly from the equations of motion. Recent mathematical developments have established connections between turbulence phenomenology and analytical properties of the Euler and Navier–Stokes equations [15, 5, 12, 19].

A first-principles analytical derivation is available for a more restricted class of problems, most notably passive-scalar turbulence in the Kraichnan model with a Gaussian velocity field that is delta-correlated in time [22, 18, 8, 38, 36, 4, 37, 14, 3]. Because the scalar equation is linear and the advecting velocity is Gaussian and delta-correlated in time, the hierarchy of equal-time scalar correlation functions closes. Anomalous scaling can then be related to homogeneous solutions, or zero modes, of the corresponding differential operators. This mechanism establishes a direct connection between anomalous exponents and the equations of motion. Its extension to nonlinear turbulence is obstructed by the unclosed Hopf hierarchy of correlation functions.

A different framework has recently been developed using a hidden scaling symmetry of turbulent dynamics in suitably rescaled variables [27, 29, 33]. The rescaling combines a change of amplitudes with a state-dependent transformation of time, thereby removing the explicit dependence on the observation scale from the inertial-range equations. The resulting universal dynamics is invariant under a change of the reference scale. Within this formulation, the statistically restored hidden symmetry induces a Perron–Frobenius description of multiplier statistics [29, 30, 6], reducing the determination of anomalous scaling exponents to an eigenvalue problem.

A perturbative realization of this idea was recently obtained for a random shell model of turbulent convection [34]. The introduction of a white-noise component made the leading-order rescaled dynamics Gaussian and allowed the stationary statistics to be calculated systematically using Gaussian calculus. Since that model is linear, however, its correlation functions also admit a closed zero-mode description. The central question is therefore whether the hidden-symmetry perturbation theory can be extended to nonlinear cascade dynamics, for which the hierarchy of correlation functions does not close.

In the present work, we develop such an extension for a stochastic modification of the dyadic

shell model. The model is nonlinear and possesses the basic structural properties associated with an energy cascade: nearest-neighbor transfer in scale space, conservation of energy by the ideal dynamics, and an exact scaling symmetry. The stochastic component is introduced into the shell-to-shell transfer processes in a way that preserves these properties. It serves primarily as an analytical device, analogous to the white-in-time random velocity in the Kraichnan model [22, 39], providing a perturbative setting in which the intermittent state emerges continuously from the deterministic Kolmogorov solution.

We first derive the exact rescaled dynamics in the inertial interval, including the drift correction generated by the state-dependent stochastic time change. These equations possess an exact hidden symmetry corresponding to a shift of the reference shell. We then introduce the multiplier variables, which connect the stationary statistics of the rescaled dynamics with the structure functions defined in the original shell variables and time. The multipliers provide a local statistical description in shell space, leading naturally to a Perron–Frobenius formulation in which the dominant eigenvalue for each order p determines the corresponding anomalous scaling exponent. Solving these problems perturbatively yields explicit analytical expressions for the anomalous exponents, which are independently verified by direct numerical simulations of the full stochastic model. The present analysis demonstrates that hidden symmetry provides a systematic analytical route from the governing equations to anomalous scaling in a genuinely nonlinear cascade model.

The paper is organized as follows. Section 2 introduces the stochastic shell model and discusses its conservation properties. Section 3 derives the rescaled inertial-range dynamics and its hidden symmetry. Sections 4–8 develop the perturbation theory, determine the stationary statistics of the rescaled variables and multipliers, and derive the anomalous scaling exponents. Section 9 summarizes the results and discusses future directions. The appendices contain the derivation of the stochastic time change, the Stratonovich–Itô conversion, and the explicit perturbation coefficients used throughout the analysis.

2 Model

2.1 Equations of motion

We consider a stochastic shell model describing energy transfer across a sequence of shells indexed by $n = 1, \dots, N$. The shell wavenumbers are defined by $k_n = \lambda^n$, where $\lambda > 1$ is the shell-spacing parameter (typically $\lambda = 2$). Each shell is associated with a real-valued amplitude $u_n(t)$, representing the characteristic velocity fluctuation at the scale $\ell_n = k_n^{-1}$.

The shell amplitudes evolve at the interior shells $n = 2, \dots, N - 1$ according to the

Stratonovich stochastic differential equations

$$du_n = u_{n-1} \circ dF_{n-1} - u_{n+1} \circ dF_n. \quad (1)$$

This equation describes the local transfer dynamics across scales. The first term represents the influx into shell n from the neighboring larger-scale shell $n-1$, while the second term represents the corresponding outflux toward the smaller-scale shell $n+1$. Such nearest-neighbor coupling provides a simplified representation of the scale-local interactions responsible for the turbulent cascade in the Navier–Stokes equations.

We define the stochastic transfer process $F_n(t)$ by

$$dF_n = k_n u_n dt + \varepsilon k_n^{1/2} |u_n|^{1/2} \circ dw_n, \quad (2)$$

where $w_n(t)$ are independent standard Wiener processes. The deterministic term represents the mean transfer rate associated with the eddy-turnover time $\tau_n \sim k_n^{-1} |u_n|^{-1}$. The stochastic term models intermittent fluctuations of the cascade activity. Its amplitude is chosen so that the fluctuations over one turnover time are of relative magnitude $O(\varepsilon)$ compared with the mean transfer, independently of scale.

At the first shell, we impose the large-scale forcing through

$$du_1 = dt - u_2 \circ dF_1, \quad (3)$$

where the term dt represents a constant forcing. For the last shell N , we choose the equation in the form

$$du_N = u_{N-1} \circ dF_{N-1} - D k_N |u_N| u_N dt, \quad (4)$$

where $D > 0$ is a dissipation coefficient. The last term acts as an energy sink at the smallest resolved scale, playing a role analogous to subgrid-scale dissipation in large-eddy simulations. One may alternatively employ a conventional viscous dissipation term $-\nu k_n^2 u_n dt$ acting on all shells. Since our analysis focuses on inertial-range dynamics, the results are insensitive to the particular form of the small-scale dissipation. We therefore adopt (4), which efficiently removes energy at the end of the cascade while minimizing the influence of the dissipation range on the inertial interval.

For $\varepsilon = 0$, Eq. (2) reduces to $dF_n = k_n u_n dt$, and the stochastic shell model (1) recovers the classical dyadic (Desnyansky–Novikov) shell model [10]. The noise amplitude in (2) is chosen so that the stochastic term has the same scaling dimension as the deterministic transfer term, preserving the scaling symmetry of the ideal dynamics. This is analogous to the construction of Kraichnan-type models [22, 39], where stochastic forcing is introduced in a manner consistent with the underlying scaling symmetries.

2.2 Energy balance

Defining the shell energy by $E_n = u_n^2$ and using the Stratonovich chain rule, we obtain

$$dE_n = d\Pi_{n-1} - d\Pi_n, \quad n = 2, \dots, N-1, \quad (5)$$

where the stochastic energy flux through the interface between shells n and $n+1$ is

$$d\Pi_n = 2u_n u_{n+1} \circ dF_n = 2k_n u_n^2 u_{n+1} dt + 2\varepsilon k_n^{1/2} |u_n|^{1/2} u_n u_{n+1} \circ dw_n. \quad (6)$$

For the first and last shells, Eqs. (3) and (4) yield

$$\begin{aligned} dE_1 &= 2u_1 dt - d\Pi_1, \\ dE_N &= d\Pi_{N-1} - 2Dk_N |u_N|^3 dt. \end{aligned} \quad (7)$$

Defining the total energy $E = \sum_{n=1}^N u_n^2$ and summing over all shells, the internal fluxes cancel telescopically, yielding

$$dE = 2u_1 dt - 2Dk_N |u_N|^3 dt. \quad (8)$$

Equation (8) shows that the stochastic transfer terms preserve the conservative inter-shell transfer of energy. The inter-shell fluxes merely redistribute energy among the shells and therefore cancel in the total energy balance. The total energy changes only through the deterministic large-scale forcing and the dissipative sink at the smallest resolved scale.

2.3 Anomalous scaling in the inertial interval

For large N , the forcing and dissipation scales are widely separated: forcing acts at the first shell $n = 1$, while dissipation acts at the last shell $n = N$. Between these regions lies a broad inertial interval,

$$1 \ll k_n \ll k_N, \quad (9)$$

where the dynamics is governed by scale-to-scale transfer. Since interactions are local in shell space and energy is exchanged conservatively between neighboring shells, the stationary cascade is characterized by a shell-independent mean energy flux, analogous to the constant-flux regime of hydrodynamic turbulence. Figure 1 illustrates a typical realization of the dynamics for $N = 24$ and $\varepsilon = 0.3$; see the caption for numerical details. The dynamics exhibits intermittent bursts, characteristic of turbulent shell models [26].

The statistical properties of the cascade are characterized by the structure functions

$$S_p(n) = \langle |u_n|^p \rangle_t = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T |u_n(t)|^p dt, \quad (10)$$

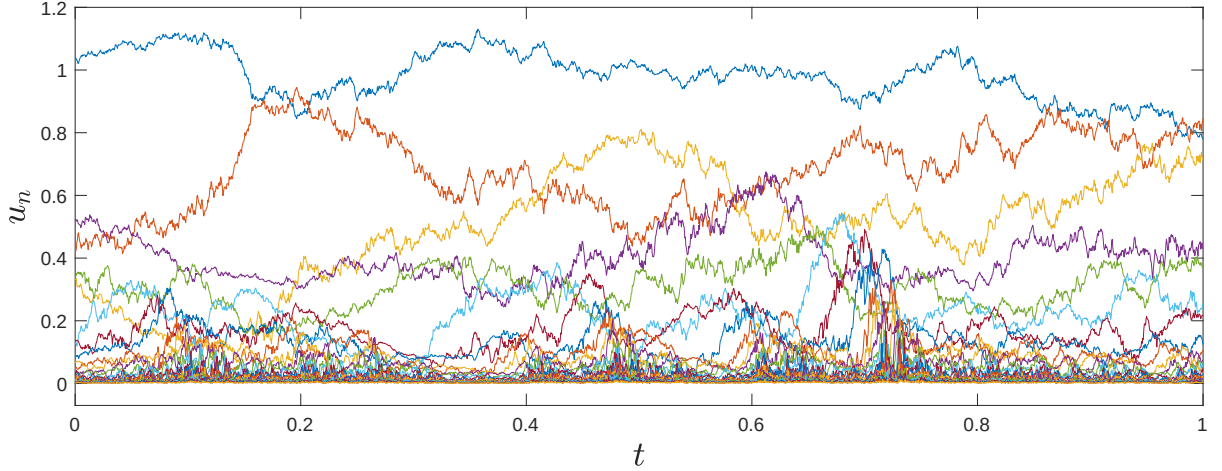


Figure 1: Typical intermittent dynamics of the shell variables $u_n(t)$, $n = 1, \dots, N$, for $\varepsilon = 0.3$. In all numerical simulations presented in this paper, we used $\lambda = 2$ and $D = 2^{-1/3}$. The Itô equations (see Appendix A.1) were integrated using the Euler–Maruyama method. The dissipation term was implemented through a low-pass filter. The time step was $\Delta t = 0.02 k_N^{-2/3}$, where $k_N^{-2/3}$ is the K41 estimate of the turnover time at the smallest resolved scale. Statistical averages were computed over a time interval of length 10^4 after discarding the initial transient.

where $\langle \cdot \rangle_t$ denotes the long-time average. In the inertial interval, the structure functions exhibit power-law scaling,

$$S_p(n) \propto k_n^{-\zeta_p}, \quad 1 \ll k_n \ll k_N, \quad (11)$$

with scaling exponents ζ_p ; see Fig. 2. The hallmark of intermittency is the nonlinear dependence of ζ_p on the moment order p .

For $\varepsilon = 0$ and $D = \lambda^{-1/3}$, the model admits the Kolmogorov stationary solution

$$u_n(t) \equiv k_n^{-1/3}. \quad (12)$$

This solution corresponds to a constant energy flux through the shells and yields the K41 prediction $\zeta_p = p/3$. For $\varepsilon > 0$, stochastic transfer fluctuations generate intermittency and lead to anomalous exponents $\zeta_p \neq p/3$.

The following sections develop an analytical theory for the inertial range. Its principal result is the perturbative expansion

$$\zeta_p = \frac{p}{3} + \varepsilon^2 \frac{p(p-3)}{6 \ln \lambda} (c_1 - \lambda^{1/3} - \lambda^{-1/3}) + O(\varepsilon^4), \quad (13)$$

valid for all orders $p \in \mathbb{R}$ for which the corresponding structure functions are finite. Here the coefficient $c_1(\lambda)$ is determined by an explicit linear system of equations. In particular,

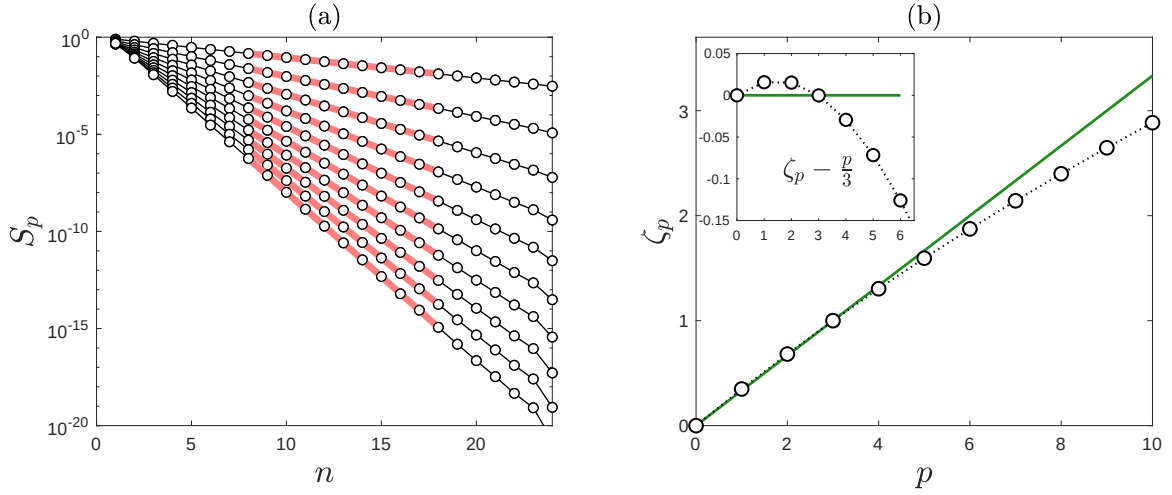


Figure 2: (a) Structure functions $S_p(n)$, $p = 1, \dots, 10$, obtained from numerical simulations (logarithmic vertical scale) for $\varepsilon = 0.3$. The red lines show the power-law fits (11) over the inertial-range shells $n = 8, \dots, 18$. (b) Estimated anomalous scaling exponents ζ_p , compared with the K41 prediction $p/3$ (solid line); error bars are omitted because they are smaller than the symbol size. The inset shows the deviations $\zeta_p - p/3$.

the perturbative correction vanishes at $p = 0$ and $p = 3$, consistently with $\zeta_0 = 0$ and the constant-flux value $\zeta_3 = 1$ [16].

Beyond this explicit result, the analysis provides a first-principles framework for anomalous scaling based on the hidden scaling symmetry of the equations of motion.

3 Hidden scaling symmetry in the inertial interval

3.1 Scaling symmetries of the ideal shell model

Dynamics in the inertial interval is governed by the ideal system

$$du_n = u_{n-1} \circ dF_{n-1} - u_{n+1} \circ dF_n. \quad (14)$$

This system possesses a family of space-time scaling symmetries

$$t, u_n, w_n \mapsto \lambda^{1-h}t, \lambda^h u_{n+1}, \lambda^{(1-h)/2} w_{n+1}, \quad (15)$$

parameterized by the Hölder exponent $h \in \mathbb{R}$. Equivalently, the transformed variables are defined by

$$u_n(t) \mapsto \lambda^h u_{n+1}\left(\frac{t}{\lambda^{1-h}}\right), \quad w_n(t) \mapsto \lambda^{(1-h)/2} w_{n+1}\left(\frac{t}{\lambda^{1-h}}\right). \quad (16)$$

By the Brownian scaling property, the transformed processes remain independent standard Wiener processes. Substituting (16) into (14) and (2), one verifies directly that the transformed variables satisfy the same equations.

A self-similar cascade characterized by a single Hölder exponent h would be statistically invariant under the corresponding transformation (15), implying the scaling exponents $\zeta_p = hp$ in Eqs. (10) and (11). In particular, the Kolmogorov stationary state $u_n = k_n^{-1/3}$ corresponds to $h = 1/3$. Intermittency, however, leads to anomalous exponents ζ_p that are nonlinear functions of p , so that no single exponent h governs the scaling of all moments. Consequently, the inertial-range statistics is not invariant under any transformation from the family (15), reflecting the breakdown of simple self-similarity.

3.2 Rescaled dynamics

Let us fix a reference shell m in the inertial interval. We introduce the rescaled variables

$$U_i = \frac{u_{m+i}}{|u_m|}, \quad d\tau = k_m |u_m| dt. \quad (17)$$

The rescaled time τ is measured in units of the instantaneous turnover time of the reference shell m . On every maximal interval where $u_m \neq 0$, the transformation (17) is well defined and the change of time variable is strictly monotone. By definition,

$$U_0(\tau) = \text{sgn}(u_m) \equiv \sigma_m, \quad \sigma_m \in \{-1, 1\}, \quad (18)$$

throughout each such interval. Applying the ordinary Stratonovich chain rule to $U_i = u_{m+i}/|u_m|$ and using Eqs. (14) and (17), we obtain (see Appendix A.2 for the derivation):

$$dU_i = U_{i-1} \circ dG_{i-1} - U_{i+1} \circ dG_i - U_i U_0 (U_{-1} \circ dG_{-1} - U_1 \circ dG_0), \quad (19)$$

where $dG_i = dF_{m+i}$. Since the time change is state dependent, expressing the transfer processes in terms of the rescaled time τ introduces an additional drift term. The resulting equation is (see Appendix A.2 for the derivation):

$$dG_i = \left[\lambda^i U_i + \frac{\varepsilon^2}{4} U_0 (\lambda^{-1} U_{-1} |U_{-1}| \delta_{i,-1} - U_1 \delta_{i0}) \right] d\tau + \varepsilon \lambda^{i/2} |U_i|^{1/2} \circ dW_i, \quad (20)$$

where

$$dW_i = (k_m |u_m|)^{1/2} dw_{m+i}. \quad (21)$$

Since

$$\langle dW_i dW_j \rangle = \delta_{ij} d\tau, \quad (22)$$

the processes $W_i(\tau)$ are independent standard Wiener processes with respect to the rescaled time τ .

The rescaled dynamics defined by Eqs. (19) and (20) depends only on the shell offsets relative to the reference shell and contains no explicit dependence on the shell number m . This universality is the origin of the hidden symmetry introduced below.

We remark that the rescaling can alternatively be formulated in terms of strictly positive local shell amplitudes [27, 30], thereby avoiding vanishing denominators in (17). We do not pursue this approach here, since it leads to a considerably more technical analysis without changing the perturbative results.

3.3 Hidden symmetry

The hidden symmetry follows from shifting the reference shell in the rescaling (17) from m to $m + 1$. We denote by primes the rescaled variables associated with the new reference shell. This shift induces the transformation (see Appendix A.2)

$$\tau \mapsto \tau', \quad U_i \mapsto U'_i, \quad dW_i \mapsto dW'_i, \quad (23)$$

defined by

$$d\tau' = \lambda |U_1| d\tau, \quad U'_i = \frac{U_{i+1}}{|U_1|}, \quad dW'_i = (\lambda |U_1|)^{1/2} dW_{i+1}. \quad (24)$$

Here $W'_i(\tau')$ are independent standard Wiener processes with respect to τ' . Since the rescaled equations hold for an arbitrary reference shell, the transformation (24) leaves the rescaled system (19) and (20) invariant.

We refer to (24) as the hidden symmetry. Unlike the family of scaling symmetries (15), it does not depend on a prescribed Hölder exponent h . The normalization by the local shell amplitude and turnover time removes the explicit scaling factors associated with any prescribed self-similar scaling, revealing a symmetry intrinsic to the rescaled cascade dynamics.

Although intermittency breaks the simple self-similarity associated with any fixed exponent h , numerical studies of both shell models [27, 30] and full turbulence models [33, 25] indicate that the hidden symmetry is statistically restored in the inertial interval. In the present work, we exploit this statistical symmetry to derive the inertial-range scaling exponents perturbatively.

4 Perturbation formalism for the rescaled system

In this section we summarize the main steps of the perturbative analysis in the inertial interval. The central idea is that the stationary Fokker–Planck equation for the rescaled system can be solved under the assumption of statistically restored hidden symmetry. This symmetry effectively replaces the boundary conditions at large and small scales by relating the statistics at neighboring scales. As a result, the inertial-range solution can be constructed without introducing explicit infrared and ultraviolet cutoffs.

4.1 Expansion of the rescaled equations

In the perturbative analysis, we consider the rescaled system (19) and (20) with the shell index extending over all integers, $i \in \mathbb{Z}$. This corresponds to the asymptotic limit of infinite scale separation between forcing and dissipation. Throughout this analysis, infinite-dimensional probability densities and their expansions are understood through finite-dimensional marginals, with the infinite inertial interval treated as an asymptotic limit.

For $\varepsilon = 0$, the deterministic rescaled system admits the stationary solution

$$U_i(\tau) \equiv \lambda^{-i/3}, \quad (25)$$

which corresponds to the Kolmogorov solution $u_n = k_n^{-1/3}$. This stationary solution is invariant under the hidden symmetry (24). We therefore seek solutions for small $\varepsilon > 0$ in the form

$$U_i(\tau) = \lambda^{-i/3} (1 + \varepsilon Z_i(\tau)), \quad (26)$$

where Z_i denotes fluctuations about the Kolmogorov solution. We restrict the perturbative analysis to the positive branch $U_i > 0$. Since $U_0 = \sigma_m = 1$ on this branch, the normalization condition (18) yields

$$Z_0(\tau) \equiv 0. \quad (27)$$

As shown later, the leading-order fluctuations Z_i are Gaussian. Since a sign change of $U_i(\tau)$ in Eq. (26) requires a fluctuation $Z_i(\tau)$ of order ε^{-1} , its probability is exponentially small in ε^{-2} , and hence smaller than every algebraic order in ε . Consequently, sign changes do not contribute to any finite order of the perturbation expansion.

For the perturbative construction of the stationary density, we use the Itô form of the rescaled dynamics, which is naturally associated with the Markov generator and the Fokker–Planck equation. Substituting the expansion (26) into Eqs. (19) and (20), we obtain the Itô stochastic system

$$dZ_i = a_i(Z; \varepsilon) d\tau + \sum_j B_{ij}(Z; \varepsilon) dW_j, \quad i \neq 0. \quad (28)$$

The Itô drift $a = (a_i)$ and the noise matrix $B = (B_{ij})$ admit the expansions

$$a(Z; \varepsilon) = A^{(0)} Z + \sum_{k \geq 1} \varepsilon^k a^{(k)}(Z), \quad B(Z; \varepsilon) = B^{(0)} + \sum_{k \geq 1} \varepsilon^k B^{(k)}(Z). \quad (29)$$

The explicit coefficients required below are collected in Appendix A.3. We also introduce the diffusion matrix

$$D(Z; \varepsilon) = B(Z; \varepsilon) B(Z; \varepsilon)^T = D^{(0)} + \sum_{k \geq 1} \varepsilon^k D^{(k)}(Z), \quad (30)$$

whose expansion is determined directly by that of B .

4.2 Expansion of the statistics

At order ε^0 , Eq. (28) reduces to the Ornstein–Uhlenbeck process

$$dZ = A^{(0)} Z d\tau + B^{(0)} dW. \quad (31)$$

Its stationary state is Gaussian, with covariance matrix C and probability density

$$P_0(z) = \mathcal{N}_C \exp\left(-\frac{1}{2} z^T C^{-1} z\right), \quad (32)$$

where $\mathcal{N}_C$ is the normalization constant and z denotes a realization of Z . The covariance matrix satisfies the Lyapunov equation

$$A^{(0)} C + C (A^{(0)})^T + D^{(0)} = 0, \quad D^{(0)} = B^{(0)} (B^{(0)})^T. \quad (33)$$

We seek the stationary density for $\varepsilon > 0$ in the form

$$P(z; \varepsilon) = P_0(z) R(z; \varepsilon), \quad (34)$$

where

$$R(z; \varepsilon) = 1 + \varepsilon r^{(1)}(z) + \varepsilon^2 r^{(2)}(z) + O(\varepsilon^3). \quad (35)$$

Normalization of $P(z; \varepsilon)$ order by order is enforced by

$$\langle r^{(k)} \rangle_0 = \int r^{(k)}(z) P_0(z) dz = 0, \quad k \geq 1, \quad (36)$$

where $\langle \cdot \rangle_0$ denotes expectation with respect to the Gaussian stationary state P_0 .

The stationary density satisfies the Fokker–Planck equation

$$-\sum_i \partial_{z_i} (a_i P) + \frac{1}{2} \sum_{i,j} \partial_{z_i} \partial_{z_j} (D_{ij} P) = 0. \quad (37)$$

Substituting the expansions of a , D , and R , and collecting equal powers of ε , yields a hierarchy of linear equations for the correction functions $r^{(k)}$. For our purposes, the full functions are not required: we will only need the first-order moments $\langle z_i r^{(1)}(z) \rangle_0$. An explicit calculation of the complete first-order correction $r^{(1)}$ for a stochastic shell model of convection was carried out in Ref. [34].

4.3 Statistically restored hidden symmetry

The stationary Fokker–Planck equation alone does not determine the statistics in the inertial interval. Its solution generally depends on boundary conditions associated with the infrared

forcing scale and the ultraviolet dissipation scale. We now show that the statistically restored hidden symmetry provides the missing closure condition. It replaces the infrared and ultraviolet boundary conditions by local relations between neighboring scales, thereby selecting an inertial-interval solution without the explicit introduction of forcing and dissipation cutoffs.

Substituting the perturbation expansions

$$U_i = \lambda^{-i/3}(1 + \varepsilon Z_i), \quad U'_i = \lambda^{-i/3}(1 + \varepsilon Z'_i),$$

into the hidden symmetry transformation (24), we obtain

$$d\tau' = \lambda^{2/3}(1 + \varepsilon Z_1) d\tau, \quad Z'_i = \frac{Z_{i+1} - Z_1}{1 + \varepsilon Z_1}. \quad (38)$$

The state-dependent time reparametrization reweights the stationary measure $P(z; \varepsilon) dz$ by the local rate $\lambda^{2/3}(1 + \varepsilon z_1)$. After applying the state transformation, the resulting stationary probability measure satisfies

$$P'(z'; \varepsilon) dz' = \frac{(1 + \varepsilon z_1) P(z; \varepsilon) dz}{\langle 1 + \varepsilon z_1 \rangle_\varepsilon}, \quad (39)$$

where $\langle \cdot \rangle_\varepsilon$ denotes expectation with respect to the stationary density $P(z; \varepsilon)$. Here, the constant factor $\lambda^{2/3}$ cancels upon normalization. The transformation law (39) is the statistical counterpart of the hidden symmetry of the rescaled system (28). Because the hidden symmetry is an exact symmetry of the stochastic dynamics, it maps stationary solutions of the Fokker–Planck equation (37) into stationary solutions of the same equation. The hidden symmetry is statistically restored when the transformed stationary probability measure coincides with the original one:

$$P' = P. \quad (40)$$

An equivalent formulation is obtained in terms of observables. For any integrable observable $F(z)$, Eqs. (39) and (40) imply

$$\langle F(z) \rangle_\varepsilon = \frac{\langle (1 + \varepsilon z_1) F(z') \rangle_\varepsilon}{\langle 1 + \varepsilon z_1 \rangle_\varepsilon}, \quad (41)$$

where $z' = (z'_i)$ is given by Eq. (38). Using the representation $P(z; \varepsilon) = P_0(z)R(z; \varepsilon)$ and writing $\langle \cdot \rangle_0$ for expectation with respect to the Gaussian density $P_0(z)$, Eq. (41) becomes

$$\langle F(z) R(z; \varepsilon) \rangle_0 = \frac{\langle (1 + \varepsilon z_1) F(z') R(z; \varepsilon) \rangle_0}{\langle (1 + \varepsilon z_1) R(z; \varepsilon) \rangle_0}. \quad (42)$$

In Eq. (42), the dependence on ε appears only inside the averaged expressions. Expanding Eq. (42) in powers of ε for polynomial observables yields a hierarchy of relations among stationary moments of the rescaled variables. For example, at leading order, setting $\varepsilon = 0$ gives

$$\langle F(z) \rangle_0 = \langle F(z') \rangle_0, \quad z'_i = z_{i+1} - z_1. \quad (43)$$

These identities provide the closure relations used below by choosing appropriate observables.

We also note an additional symmetry associated with the sign of the noise amplitude. Since the Wiener processes are invariant in law under $W_i \mapsto -W_i$, the rescaled stochastic dynamics is invariant in law under $\varepsilon \mapsto -\varepsilon$. In terms of the perturbation variables defined by Eq. (26), this transformation is accompanied by $Z_i \mapsto -Z_i$. Thus,

$$\varepsilon \mapsto -\varepsilon, \quad W_i \mapsto -W_i, \quad Z_i \mapsto -Z_i. \quad (44)$$

The corresponding stationary density satisfies

$$P(z; \varepsilon) = P(-z; -\varepsilon). \quad (45)$$

This symmetry will be used below to constrain the perturbation expansions.

5 Leading-order Gaussian solution

At leading order, the stationary statistics is Gaussian, with probability density (32). In this section, we determine the covariance matrix

$$C_{ij} = \langle z_i z_j \rangle_0. \quad (46)$$

Additionally, we compute the first-order corrections to the mean values

$$v_i = \langle z_i r^{(1)}(z) \rangle_0. \quad (47)$$

Finally, we compare the analytical predictions with numerical simulations.

5.1 Hidden symmetry of the covariance matrix

The leading-order hidden symmetry implies explicit constraints on the covariance matrix. Taking $F(z) = z_i z_j$ in Eqs. (43), we obtain

$$\langle z_i z_j \rangle_0 = \langle (z_{i+1} - z_1)(z_{j+1} - z_1) \rangle_0. \quad (48)$$

Expanding the product and using the definition (46) gives the recursion

$$C_{ij} = C_{i+1,j+1} - C_{i+1,1} - C_{1,j+1} + C_{1,1}, \quad (49)$$

for arbitrary integers i and j .

We introduce the notation

$$c_0 = 0, \quad c_i = C_{i1} = C_{1i}, \quad i \geq 1. \quad (50)$$

By repeated application of Eq. (49), every covariance C_{ij} can be expressed as a linear combination of the coefficients c_i . Rather than writing the general expression, we record below only the relations needed in the subsequent analysis. Setting $(i, j) \mapsto (i-1, 1)$, $(i, j) \mapsto (i, -1)$, or $(i, j) \mapsto (-1, -1)$ in Eq. (49) and using the normalization $C_{i0} = 0$ yields

$$C_{i2} = c_{i-1} + c_i + c_2 - c_1, \quad C_{i,-1} = -c_{i+1} + c_1, \quad C_{-1,-1} = c_1, \quad i \geq 1. \quad (51)$$

Applying Eq. (49) recursively for $i \geq 1$, we obtain the following relations for the diagonal and first off-diagonal entries:

$$C_{ii} = 2 \sum_{k=1}^i c_k - i c_1, \quad C_{i,i-1} = 2 \sum_{k=1}^i c_k - c_i - i c_1, \quad i \geq 1. \quad (52)$$

5.2 Covariance from the Lyapunov equation

The stationary Gaussian distribution (32) of the leading-order Ornstein–Uhlenbeck dynamics (31) is determined by the covariance matrix C , which satisfies the Lyapunov equation (33). Using the explicit expressions for $A^{(0)}$ and $B^{(0)}$ from Appendix A.3, together with Eqs. (50) and (51), we evaluate its $(i, 1)$ components for $i \geq 1$:

$$(A^{(0)}C)_{i1} = \lambda^{(2i-1)/3} (2c_{i-1} - c_i - c_{i+1}) - \lambda^{-1/3} (c_1 - 2c_2), \quad (53)$$

$$(C(A^{(0)})^T)_{i1} = -\lambda^{1/3} c_{i-1} + 2\lambda^{-1/3} c_{i+1} + (-2\lambda^{1/3} + \lambda^{-1/3}) c_i - \lambda^{1/3} c_2 + (\lambda^{1/3} - 2\lambda^{-1/3}) c_1, \quad (54)$$

$$D_{i1}^{(0)} = 2 + \lambda^{-2/3} + (2 + \lambda^{2/3}) \delta_{i1} - \lambda^{2/3} \delta_{i2}. \quad (55)$$

Substituting Eqs. (53)–(55) into the $(i, 1)$ component of the Lyapunov equation gives

$$\begin{aligned} 0 = & (2\lambda^{(2i-1)/3} - \lambda^{1/3}) c_{i-1} - (\lambda^{(2i-1)/3} + 2\lambda^{1/3} - \lambda^{-1/3}) c_i \\ & + (-\lambda^{(2i-1)/3} + 2\lambda^{-1/3}) c_{i+1} + (\lambda^{1/3} - 3\lambda^{-1/3}) c_1 + (2\lambda^{-1/3} - \lambda^{1/3}) c_2 \\ & + 2 + \lambda^{-2/3} + (2 + \lambda^{2/3}) \delta_{i1} - \lambda^{2/3} \delta_{i2}, \quad i \geq 1. \end{aligned} \quad (56)$$

Because the leading-order Lyapunov equation is invariant under the hidden-symmetry transformation, its remaining components are generated from the $(i, 1)$ components by the recursion (49). Thus, it is sufficient to solve Eq. (56).

For $i \rightarrow \infty$, dividing Eq. (56) by $\lambda^{(2i-1)/3}$ and neglecting the vanishing terms yields

$$2c_{i-1} - c_i - c_{i+1} \simeq 0. \quad (57)$$

Seeking solutions of the form $c_i = r^i$ yields the characteristic equation $r^2 + r - 2 = 0$, whose roots are $r_1 = 1$ and $r_2 = -2$. The boundedness of (c_i) excludes the exponentially growing mode $(-2)^i$. We therefore impose the asymptotic condition

$$\lim_{i \rightarrow \infty} c_i = c_\infty, \quad (58)$$

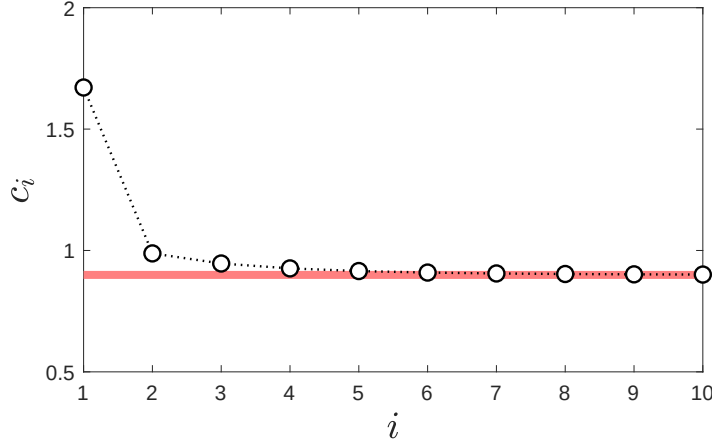


Figure 3: Theoretical covariances $c_i = \langle z_1 z_i \rangle_0$ for $\lambda = 2$. Circles show the numerical solution of the truncated recurrence (56) with $i_{\max} = 30$. The red line indicates the asymptotic value c_∞ given by Eq. (60).

where c_∞ is a finite constant. The asymptotic condition (58) closes the infinite linear system (56), thereby selecting the covariance matrix C .

Multiplying Eq. (56) by $\lambda^{-(2i-1)/3}$ and summing over $i \geq 1$, the sums telescope. Using $c_0 = 0$ and the asymptotic condition (58), we obtain

$$c_\infty = \frac{\lambda^{1/3} (\lambda^{2/3} + 2) - 2c_1 + (2 - \lambda^{2/3}) c_2}{3(\lambda^{2/3} - 1)}. \quad (59)$$

Combining this relation with Eq. (56) at $i = 1$ gives the equivalent expression

$$c_\infty = \frac{c_1}{3} + \frac{1}{6} (\lambda^{1/3} + \lambda^{-1/3}). \quad (60)$$

System (56) is linear and can be solved numerically by truncating it to $1 \leq i \leq i_{\max}$ and imposing the asymptotic boundary condition (58) in the form $c_{i_{\max}+1} = c_{i_{\max}}$. The solution converges rapidly as $i_{\max}$ is increased. The circles in Fig. 3 show the solution obtained with $i_{\max} = 30$, confirming the convergence to the asymptotic value c_∞ given by Eq. (60) (red line).

The finite limit (58) admits a simple physical interpretation. For widely separated shells, the larger-scale shell evolves much more slowly than the smaller-scale one and therefore acts as a slowly varying modulation of the cascade. This modulation is transmitted across the inertial interval, so its influence does not vanish with shell separation, resulting in the nonzero asymptotic covariance $c_i \rightarrow c_\infty$.

5.3 Moments of the first-order correction

The moments introduced in Eq. (47) determine the first-order correction to the mean:

$$\langle z_i \rangle_\varepsilon = \langle z_i R(z; \varepsilon) \rangle_0 = \varepsilon \langle z_i r^{(1)}(z) \rangle_0 + O(\varepsilon^2) = \varepsilon v_i + O(\varepsilon^2), \quad (61)$$

where we used Eqs. (34) and (35).

Consider now the hidden-symmetry condition (42) with $F(z) = z_i$. Using Eq. (38), the transformed variables satisfy

$$(1 + \varepsilon z_1) z'_i = z_{i+1} - z_1. \quad (62)$$

Using the expansion (35) with the normalization condition (36), one can see that

$$\langle (1 + \varepsilon z_1) R(z; \varepsilon) \rangle_0 = 1 + O(\varepsilon^2). \quad (63)$$

Expanding both sides of the resulting equality (42) to first order in ε yields

$$\langle z_i r^{(1)}(z) \rangle_0 = \langle (z_{i+1} - z_1) r^{(1)}(z) \rangle_0, \quad (64)$$

or, equivalently,

$$v_i = v_{i+1} - v_1. \quad (65)$$

Since $v_0 = 0$ follows from the identity $z_0 \equiv 0$, the recursion is solved by

$$v_i = i v_1, \quad i \in \mathbb{Z}. \quad (66)$$

The remaining constant v_1 can be determined from the stationarity condition for the first moment,

$$\langle a_1(z; \varepsilon) \rangle_\varepsilon = \langle a_1(z; \varepsilon) R(z; \varepsilon) \rangle_0 = 0. \quad (67)$$

Using Eqs. (29) and (35), we obtain to first order in ε :

$$\sum_j A_{1j}^{(0)} \langle z_j r^{(1)}(z) \rangle_0 + \langle a_1^{(1)}(z) \rangle_0 = 0. \quad (68)$$

Using Eqs. (47) and (66), together with the explicit form of $A^{(0)}$ in Eq. (158), the first term evaluates to

$$\sum_j A_{1j}^{(0)} \langle z_j r^{(1)}(z) \rangle_0 = \sum_j A_{1j}^{(0)} v_j = 3 (\lambda^{-1/3} - \lambda^{1/3}) v_1. \quad (69)$$

Using Eq. (159) at $i = 1$ with Eqs. (50) and (51), the second term evaluates to

$$\langle a_1^{(1)}(z) \rangle_0 = (2\lambda^{-1/3} - \lambda^{1/3}) c_2 - 2\lambda^{-1/3} c_1 + 2 - \frac{1}{2} \lambda^{2/3} + \frac{3}{2} \lambda^{-2/3}. \quad (70)$$

Using Eq. (56) at $i = 1$, this expression simplifies to

$$\langle a_1^{(1)}(z) \rangle_0 = (\lambda^{1/3} - \lambda^{-1/3}) (c_1 - \lambda^{1/3} - \lambda^{-1/3}). \quad (71)$$

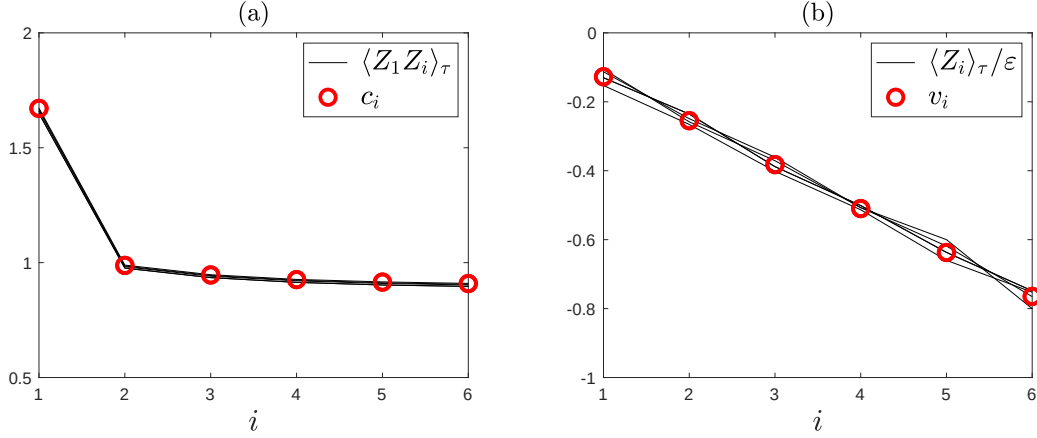


Figure 4: (a) Stationary covariances $\langle Z_1 Z_i \rangle_\tau$ obtained from numerical simulations (thin black lines) compared with the theoretical predictions c_i (red circles). Four almost indistinguishable curves correspond to the reference shells $m = 9, \dots, 12$ within the inertial interval. (b) Similar comparison for the normalized expectations $\langle Z_i \rangle_\tau / \varepsilon$ (thin black lines) and their theoretical predictions v_i (red circles). Calculations are performed for $\varepsilon = 0.05$.

Substituting Eqs. (69) and (71) into Eq. (68), we obtain

$$v_1 = \frac{1}{3} (c_1 - \lambda^{1/3} - \lambda^{-1/3}). \quad (72)$$

Combining Eqs. (66) and (72) yields

$$v_i = \frac{i}{3} (c_1 - \lambda^{1/3} - \lambda^{-1/3}). \quad (73)$$

5.4 Comparison with numerical simulations

We now compare theoretical predictions with direct numerical simulations. Assuming ergodicity, the stationary expectation $\langle \cdot \rangle_\varepsilon$ associated with the rescaled dynamics is evaluated numerically by long-time averaging $\langle \cdot \rangle_\tau$ with respect to the rescaled time τ . The thin black lines in Fig. 4(a) show the stationary covariances $\langle Z_1 Z_i \rangle_\tau$ obtained from numerical simulations with $\varepsilon = 0.05$, where $Z_i(\tau)$ is defined by Eqs. (17) and (26). Excellent agreement is observed with the theoretical covariance coefficients c_i (red circles). The thin black lines in Fig. 4(b) show the normalized expectations $\langle Z_i \rangle_\tau / \varepsilon$, where the normalization follows from Eq. (61). The statistical uncertainty is noticeably larger in this case because the expectations themselves are $O(\varepsilon)$. Nevertheless, the agreement with the theoretical predictions v_i , computed from Eq. (73), is again very good.

The covariance matrix determines all finite-dimensional Gaussian marginals. In particular,

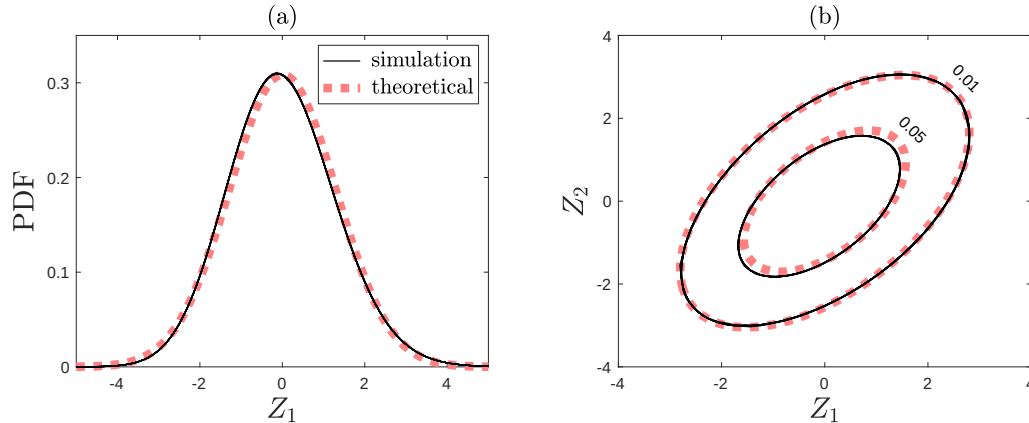


Figure 5: (a) Probability density functions obtained from numerical simulations (thin black lines) compared with the leading-order Gaussian distribution (74) (thick dotted line). Fourteen visually indistinguishable black curves correspond to the reference shells $m = 5, \dots, 18$. (b) Similar comparison for the contour levels 0.01 and 0.05 of the joint probability density of (Z_1, Z_2) . Calculations are performed for $\varepsilon = 0.05$.

to leading order,

$$Z_1 \sim \mathcal{N}(0, c_1), \quad (74)$$

while the pair (Z_1, Z_2) has the two-dimensional Gaussian distribution with covariance matrix

$$\begin{pmatrix} c_1 & c_2 \\ c_2 & 2c_2 \end{pmatrix}, \quad (75)$$

where the identity $C_{22} = 2c_2$ follows from Eq. (52). The corresponding numerical distributions are shown in Fig. 5. The numerical one- and two-dimensional probability density functions agree remarkably well with the Gaussian predictions (74) and (75). This confirms that the leading-order Gaussian approximation accurately describes the stationary statistics in the weak-noise regime.

5.5 A shortcut derivation of anomalous exponents

We first present a simpler derivation of the perturbative anomalous exponents based on the Gaussian statistics of the rescaled variables. It relies on two nontrivial assumptions: the existence of inertial-range power-law scaling and its identification with the asymptotic scaling of the moments of the rescaled variables. The Perron–Frobenius formulation developed later provides the connection between the multiplier statistics and the scaling of the original structure functions, making the present derivation a convenient analytical shortcut.

Let us assume that the moments of the rescaled variables satisfy the same power-law scaling as the original shell variables,

$$\langle U_i^p \rangle_\tau \propto k_i^{-\zeta_p}, \quad k_i = \lambda^i, \quad (76)$$

for sufficiently large shell numbers i in the inertial interval. Here $\langle \cdot \rangle_\tau$ denotes the stationary average with respect to the rescaled dynamics for a fixed reference shell m . This assumption is motivated by the fact that, for large i , the reference shell m evolves much more slowly than shell $m + i$, so that the rescaling introduces only a slowly varying prefactor and is therefore expected not to affect the scaling exponent.

Assuming ergodicity, we replace the rescaled-time average by the stationary average with respect to the density $P(z; \varepsilon)$. Using Eqs. (34), (35), and (26), we obtain

$$\begin{aligned} \langle U_i^p \rangle_\tau &= \lambda^{-ip/3} \langle (1 + \varepsilon z_i)^p [1 + \varepsilon r^{(1)}(z) + \varepsilon^2 r^{(2)}(z) + O(\varepsilon^3)] \rangle_0 \\ &= \lambda^{-ip/3} \left(1 + \varepsilon^2 \frac{p(p-1)}{2} \langle z_i^2 \rangle_0 + \varepsilon^2 p \langle z_i r^{(1)}(z) \rangle_0 + O(\varepsilon^4) \right), \end{aligned} \quad (77)$$

where we expanded the averaged expression in ε and used $\langle z_i \rangle_0 = \langle r^{(1)} \rangle_0 = \langle r^{(2)} \rangle_0 = 0$. The absence of odd powers in the last expression follows from the invariance in law under $\varepsilon \mapsto -\varepsilon$.

Using Eq. (52), the asymptotic condition (58), and the relation (60), we obtain

$$\begin{aligned} \langle z_i^2 \rangle_0 &= C_{ii} = 2 \sum_{k=1}^i c_k - i c_1 = i(2c_\infty - c_1) + o(i) \\ &= -\frac{i}{3} (c_1 - \lambda^{1/3} - \lambda^{-1/3}) + o(i). \end{aligned} \quad (78)$$

The remaining term is obtained from Eq. (73):

$$\langle z_i r^{(1)}(z) \rangle_0 = v_i = \frac{i}{3} (c_1 - \lambda^{1/3} - \lambda^{-1/3}). \quad (79)$$

Substituting these expressions into Eq. (77), we obtain

$$\langle U_i^p \rangle_\tau = \lambda^{-ip/3} \left[1 - i\varepsilon^2 \frac{p(p-3)}{6} (c_1 - \lambda^{1/3} - \lambda^{-1/3}) + i\varepsilon^2 o(1) \right]. \quad (80)$$

This asymptotic expansion agrees with the ε -expansion of the power-law (76) for

$$\zeta_p = \frac{p}{3} + \varepsilon^2 \frac{p(p-3)}{6 \ln \lambda} (c_1 - \lambda^{1/3} - \lambda^{-1/3}) + O(\varepsilon^4). \quad (81)$$

The above argument does not constitute a self-contained derivation of Eq. (81). First, the power-law behavior (76) is assumed rather than derived from the rescaled dynamics. Second, its identification with the scaling of the original shell variables has not yet been justified. Finally, the matching of the asymptotic expansion (80) with the power-law form (76) is only formal, because the perturbation expansion breaks down in the limit $i \rightarrow \infty$ at fixed ε , where the correction $i\varepsilon^2$ is no longer small. The Perron–Frobenius construction developed below addresses these issues and recovers the same perturbative exponent from the multiplier dynamics.

6 Multipliers

We now introduce the multiplier variables underlying the Perron–Frobenius formulation. In this section, we derive their perturbative statistics. The multiplier formulation is obtained by a change of variables from the rescaled system and inherits the Gaussian form of the leading-order probability distribution.

6.1 Statistics of multipliers

Our multiplier variables are defined as ratios of the rescaled variables

$$X_i = \frac{U_i}{U_{i-1}}. \quad (82)$$

Note that Eq. (17) implies $X_i = u_{m+i}/u_{m+i-1}$ in terms of the original shell variables. For the Kolmogorov solution $U_i = \lambda^{-i/3}$, all multipliers take the constant value

$$X_i = \lambda^{-1/3}. \quad (83)$$

Accordingly, we write

$$X_i = \lambda^{-1/3}(1 + \varepsilon Y_i), \quad (84)$$

where Y_i represents the fluctuations of the multipliers about the Kolmogorov value. The perturbative analysis of the rescaled variables is developed in the positive sector $U_i > 0$, which implies $X_i > 0$.

The stationary probability density of the multiplier fluctuations is expanded as

$$P(y; \varepsilon) = P_0(y) [1 + \varepsilon \rho^{(1)}(y) + \varepsilon^2 \rho^{(2)}(y) + O(\varepsilon^3)], \quad (85)$$

where $y = (y_i)$ denotes a realization of the random variable $Y = (Y_i)$. This density is obtained from the stationary density $P(z; \varepsilon)$ through the change of variables

$$y_i = \frac{z_i - z_{i-1}}{1 + \varepsilon z_{i-1}}, \quad (86)$$

which follows from Eqs. (26), (82), and (84). This distribution inherits the sign symmetry

$$P(y; \varepsilon) = P(-y; -\varepsilon), \quad (87)$$

from Eq. (45).

At leading order, Eq. (86) becomes $y_i = z_i - z_{i-1}$. Hence, $P_0(y)$ is the zero-mean Gaussian density

$$P_0(y) = \mathcal{N}_K \exp\left(-\frac{1}{2} y^T K^{-1} y\right). \quad (88)$$

The covariances are found as

$$K_{ij} = \langle y_i y_j \rangle_0 = \langle (z_i - z_{i-1})(z_j - z_{j-1}) \rangle_0 = C_{ij} - C_{i-1,j} - C_{i,j-1} + C_{i-1,j-1}. \quad (89)$$

Applying the covariance recursion (49) to the four terms in Eq. (89), the terms generated by the recursion cancel, yielding the Toeplitz property

$$K_{ij} = C_{i+1,j+1} - C_{i,j+1} - C_{i+1,j} + C_{ij} = K_{i+1,j+1}. \quad (90)$$

Together with the symmetry $K_{ij} = K_{ji}$, this implies

$$K_{ij} = \kappa_{|i-j|}, \quad (91)$$

where

$$\kappa_i = K_{1,i+1} = C_{1,i+1} - C_{0,i+1} - C_{1,i} + C_{0,i} = c_{i+1} - c_i, \quad i \geq 0, \quad (92)$$

using $C_{0i} = 0$, $C_{1i} = c_i$ for $i \geq 1$, and $c_0 = 0$. Since $\kappa_i \rightarrow 0$ as $i \rightarrow \infty$, multipliers decorrelate at large shell separations.

Higher-order corrections satisfy the normalization conditions

$$\langle \rho^{(k)}(y) \rangle_0 = 0, \quad k \geq 1. \quad (93)$$

Working in the positive sector, we have $U_0 \equiv 1$, so that $X_1 = U_1$ and hence $Y_1 = Z_1$. Expanding the resulting equality $\langle y_1 \rangle_\varepsilon = \langle z_1 \rangle_\varepsilon$ to first order in ε using Eqs. (61) and (85) yields

$$\langle y_1 \rho^{(1)}(y) \rangle_0 = v_1. \quad (94)$$

We now compare the analytical predictions with direct numerical simulations. Figure 6(a) shows the stationary multiplier correlations for $\varepsilon = 0.05$. Results obtained for different reference shells and shell offsets within the inertial interval collapse onto the theoretical covariance coefficients κ_i . In particular, their independence of the reference shell provides direct numerical evidence for the statistical restoration of hidden symmetry in the inertial interval.

Figure 6(b) compares the numerical probability densities of the multiplier fluctuations with the leading-order Gaussian prediction of standard deviation $\sqrt{\kappa_0}$. The agreement is very good throughout the inertial interval.

6.2 Hidden symmetry of multiplier statistics

Under the hidden-symmetry transformation (24), the multiplier variables transform as

$$X'_i = \frac{U'_i}{U'_{i-1}} = \frac{U_{i+1}}{U_i} = X_{i+1}. \quad (95)$$

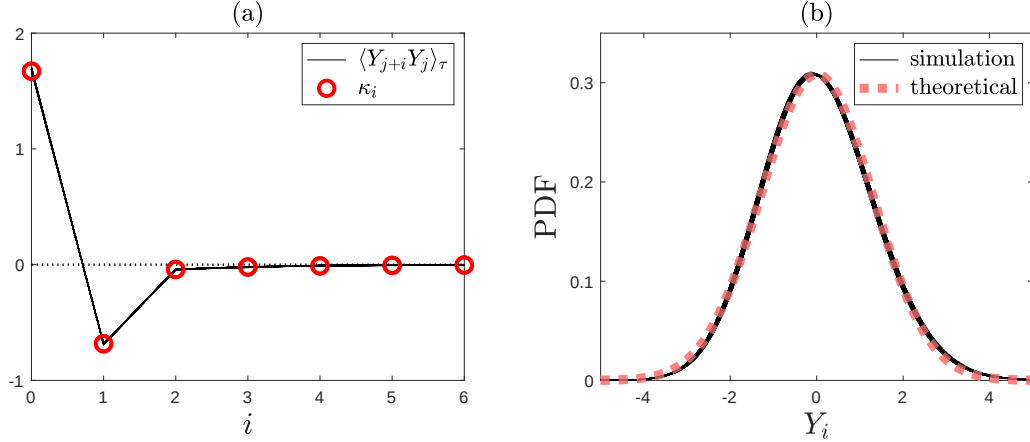


Figure 6: (a) Stationary correlations $\langle Y_{j+i} Y_j \rangle_\tau$ obtained from numerical simulations (thin black lines) compared with the theoretical covariances κ_i (red circles). The many visually indistinguishable curves correspond to different choices of i , j , and m within the inertial interval of shells $5, \dots, 18$. (b) Probability density functions of the multiplier fluctuations Y_i obtained from numerical simulations (thin black lines) for different values of i and m within the inertial interval, compared with the leading-order Gaussian prediction (thick dotted line). Calculations are performed for $\varepsilon = 0.05$.

Using Eq. (84), this reduces to a shift of the multiplier fluctuations,

$$Y'_i = Y_{i+1}. \quad (96)$$

The corresponding transformation of the rescaled time is

$$d\tau' = \lambda^{2/3} (1 + \varepsilon Y_1) d\tau. \quad (97)$$

Thus, the hidden symmetry acts on the multiplier fluctuations through the simple shift (96) of the shell index together with the state-dependent time transformation (97).

Although the transformation of the multiplier variables is a simple shift, the state-dependent time transformation modifies the stationary density. Consequently,

$$P'(y'; \varepsilon) = \frac{(1 + \varepsilon y_1) P(y; \varepsilon)}{\langle 1 + \varepsilon y_1 \rangle_\varepsilon}, \quad (98)$$

where $y'_i = y_{i+1}$ and $\langle \cdot \rangle_\varepsilon$ denotes expectation with respect to $P(y; \varepsilon)$. Statistical restoration of the hidden symmetry requires the transformed stationary density to coincide with the original one,

$$P' = P. \quad (99)$$

Substituting the perturbation expansion (85) into Eqs. (98) and (99), and comparing equal powers of ε , we obtain at leading order

$$P_0(y') = P_0(y). \quad (100)$$

This shift invariance follows directly from the Toeplitz form (91) of the covariance matrix K . At first order,

$$\rho^{(1)}(y') = \rho^{(1)}(y) + y_1, \quad (101)$$

where we used $\langle 1 + \varepsilon y_1 \rangle_\varepsilon = 1 + O(\varepsilon^2)$, following from $\langle y_1 \rangle_0 = 0$.

6.3 Marginal statistics of multipliers

This subsection collects technical relations for the marginal distributions required later in the Perron–Frobenius formulation. We denote by $P(y_\ominus; \varepsilon)$ the marginal density restricted to the variables $y_\ominus = (y_i)_{i \leq 0}$, and by $P_0(y_\ominus)$ the corresponding marginal Gaussian density. Then

$$P(y_\ominus; \varepsilon) = P_0(y_\ominus) [1 + \varepsilon \rho^{(1)}(y_\ominus) + \varepsilon^2 \rho^{(2)}(y_\ominus) + O(\varepsilon^3)], \quad (102)$$

where

$$\rho^{(k)}(y_\ominus) = \mathbb{E}_0[\rho^{(k)}(y) \mid y_\ominus]. \quad (103)$$

Similarly, for the marginal density in the variables $(y_1, y_\ominus)$,

$$P(y_1, y_\ominus; \varepsilon) = P_0(y_1, y_\ominus) [1 + \varepsilon \rho^{(1)}(y_1, y_\ominus) + \varepsilon^2 \rho^{(2)}(y_1, y_\ominus) + O(\varepsilon^3)], \quad (104)$$

where

$$\rho^{(k)}(y_1, y_\ominus) = \mathbb{E}_0[\rho^{(k)}(y) \mid y_1, y_\ominus]. \quad (105)$$

The corresponding correction terms are related by

$$\rho^{(k)}(y_\ominus) P_0(y_\ominus) = \int \rho^{(k)}(y_1, y_\ominus) P_0(y_1, y_\ominus) dy_1. \quad (106)$$

Using Eqs. (102) and (104), we obtain the expansion of the conditional probability density

$$P(y_1 | y_\ominus; \varepsilon) = \frac{P(y_1, y_\ominus; \varepsilon)}{P(y_\ominus; \varepsilon)} = \frac{P_0(y_1, y_\ominus)}{P_0(y_\ominus)} [1 + \varepsilon \sigma^{(1)}(y_1, y_\ominus) + \varepsilon^2 \sigma^{(2)}(y_1, y_\ominus) + O(\varepsilon^3)], \quad (107)$$

where

$$\sigma^{(1)}(y_1, y_\ominus) = \rho^{(1)}(y_1, y_\ominus) - \rho^{(1)}(y_\ominus), \quad (108)$$

$$\sigma^{(2)}(y_1, y_\ominus) = \rho^{(2)}(y_1, y_\ominus) - \rho^{(2)}(y_\ominus) - \rho^{(1)}(y_1, y_\ominus) \rho^{(1)}(y_\ominus) + [\rho^{(1)}(y_\ominus)]^2. \quad (109)$$

The hidden-symmetry relation (101) induces a corresponding relation for the marginal corrections. Taking the marginal of Eq. (101) in the variables $y'_\ominus = (y'_i)_{i \leq 0}$, which correspond under the shift $y'_i = y_{i+1}$ to $(y_1, y_\ominus)$, we obtain

$$\rho^{(1)}(y'_\ominus) = \rho^{(1)}(y_1, y_\ominus) + y_1. \quad (110)$$

At leading order, the Gaussian marginals satisfy

$$P_0(y'_\ominus) = P_0(y_1, y_\ominus), \quad (111)$$

which follows immediately from the shift invariance (100).

7 Structure functions as Perron–Frobenius modes

The goal of this section is to derive the structure-function scaling exponents from the rescaled formulation. The main difficulty is that structure functions are defined in terms of the original shell variables and averages with respect to the original time, whereas the analytical description is formulated in terms of the rescaled dynamics and its hidden symmetry. We establish the correspondence between these two formulations and show that the structure functions are represented as Perron–Frobenius eigenmodes associated with the multiplier formulation. The analysis in this section closely follows the corresponding Perron–Frobenius theory developed earlier for the Sabra shell model [29, 30].

7.1 Multipliers representation of structure functions

We now return to the full original system with forcing and dissipation; see Section 2. For technical reasons, it is convenient to set $u_n \equiv 1$ for the shells $n \leq 0$, which do not affect the dynamics. Then, using Eqs. (17), (82), and (84), the shell amplitude at shell m is written as the telescopic product

$$u_m = \prod_{i=1-m}^0 X_i = \lambda^{-m/3} \prod_{i=1-m}^0 (1 + \varepsilon Y_i). \quad (112)$$

As discussed in Section 6.1, the perturbative analysis is restricted to the positive sector $X_i = \lambda^{-1/3}(1 + \varepsilon Y_i) > 0$. Then the time rescaling is given by

$$dt = (k_m u_m)^{-1} d\tau = \lambda^{-2m/3} \left(\prod_{i=1-m}^0 (1 + \varepsilon Y_i)^{-1} \right) d\tau. \quad (113)$$

Substituting the expression for u_m from Eq. (112) and using Eq. (113) to express the time average in terms of the rescaled-time average, we obtain

$$S_p(m) = \langle |u_m|^p \rangle_t = \lambda^{-mp/3} \frac{\langle \prod_{i=1-m}^0 (1 + \varepsilon Y_i)^{p-1} \rangle_\tau}{\langle \prod_{i=1-m}^0 (1 + \varepsilon Y_i)^{-1} \rangle_\tau}. \quad (114)$$

From now on, we indicate the reference shell m by a superscript. For example, we write $P^{(m)}(y; \varepsilon)$ for the stationary distribution associated with reference shell m . Assuming ergodicity, the rescaled-time average $\langle \cdot \rangle_\tau$ can be identified with the stationary average over the multiplier distribution $P^{(m)}(y; \varepsilon)$. Thus, Eq. (114) can be written as

$$S_p(m) = \lambda^{-mp/3} \int Q_p^{(m)}(y_\ominus; \varepsilon) dy_\ominus, \quad (115)$$

where the density

$$Q_p^{(m)}(y_\ominus; \varepsilon) = \frac{1}{b_m} \left(\prod_{i=1-m}^0 (1 + \varepsilon y_i)^{p-1} \right) P^{(m)}(y_\ominus; \varepsilon), \quad (116)$$

and the constant

$$b_m = \int \left(\prod_{i=1-m}^0 (1 + \varepsilon y_i)^{-1} \right) P^{(m)}(y_\ominus; \varepsilon) dy_\ominus. \quad (117)$$

7.2 Recursive relation

The densities $Q_p^{(m)}$ are not themselves universal, because their definition (116) involves multipliers extending into the forcing range. Universality emerges instead through a recursive relation between the densities $Q_p^{(m)}(y_\ominus; \varepsilon)$ and $Q_p^{(m+1)}(y'_\ominus; \varepsilon)$, associated with the adjacent reference shells m and $m+1$. We first show that

$$Q_p^{(m+1)}(y'_\ominus; \varepsilon) = (1 + \varepsilon y_1)^p P^{(m)}(y_1 | y_\ominus; \varepsilon) Q_p^{(m)}(y_\ominus; \varepsilon), \quad (118)$$

where $P^{(m)}(y_1 | y_\ominus; \varepsilon)$ is the conditional probability density associated with the reference shell m , and the arguments $y_\ominus = (y_i)_{i \leq 0}$ and $y'_\ominus = (y'_i)_{i \leq 0}$ are related by $y'_i = y_{i+1}$.

Let $P^{(m+1)}(y'_\ominus; \varepsilon)$ be the stationary multiplier distribution associated with the reference shell $m+1$. Under the shift $y'_i = y_{i+1}$, the variables $y'_\ominus$ are identified with $(y_1, y_\ominus)$. The change of reference shell $m \mapsto m+1$ induces the hidden-symmetry transformation (98), with P and P' there corresponding respectively to $P^{(m)}$ and $P^{(m+1)}$ in the present notation. Marginalizing this relation to the variables $(y_1, y_\ominus)$ yields

$$P^{(m+1)}(y'_\ominus; \varepsilon) = \frac{1 + \varepsilon y_1}{\beta_m} P^{(m)}(y_1, y_\ominus; \varepsilon), \quad (119)$$

with the normalization constant

$$\beta_m = \int (1 + \varepsilon y_1) P^{(m)}(y_1, y_\ominus; \varepsilon) dy_1 dy_\ominus. \quad (120)$$

Writing Eq. (116) at the reference shell $m + 1$ and substituting Eq. (119) with $y'_i = y_{i+1}$ yields

$$\begin{aligned}
Q_p^{(m+1)}(y'_\ominus; \varepsilon) &= \frac{1}{b_{m+1}} \left(\prod_{i=-m}^0 (1 + \varepsilon y'_i)^{p-1} \right) P^{(m+1)}(y'_\ominus; \varepsilon) \\
&= \frac{1}{b_{m+1}} \left(\prod_{i=1-m}^1 (1 + \varepsilon y_i)^{p-1} \right) \frac{1 + \varepsilon y_1}{\beta_m} P^{(m)}(y_1, y_\ominus; \varepsilon) \\
&= \frac{b_m(1 + \varepsilon y_1)^p}{b_{m+1}\beta_m} \frac{1}{b_m} \left(\prod_{i=1-m}^0 (1 + \varepsilon y_i)^{p-1} \right) P^{(m)}(y_\ominus; \varepsilon) P^{(m)}(y_1|y_\ominus; \varepsilon) \\
&= \frac{b_m(1 + \varepsilon y_1)^p}{b_{m+1}\beta_m} Q_p^{(m)}(y_\ominus; \varepsilon) P^{(m)}(y_1|y_\ominus; \varepsilon). \tag{121}
\end{aligned}$$

Finally, using Eqs. (117) and (120) we express the prefactor as

$$\begin{aligned}
b_{m+1} &= \int \left(\prod_{i=-m}^0 (1 + \varepsilon y'_i)^{-1} \right) P^{(m+1)}(y'_\ominus; \varepsilon) dy'_\ominus \\
&= \int \left(\prod_{i=1-m}^1 (1 + \varepsilon y_i)^{-1} \right) \frac{1 + \varepsilon y_1}{\beta_m} P^{(m)}(y_1, y_\ominus; \varepsilon) dy_1 dy_\ominus \\
&= \frac{1}{\beta_m} \int \left(\prod_{i=1-m}^0 (1 + \varepsilon y_i)^{-1} \right) P^{(m)}(y_\ominus; \varepsilon) dy_\ominus = \frac{b_m}{\beta_m}. \tag{122}
\end{aligned}$$

Substituting (122) into (121) yields Eq. (118).

By scale locality, the conditional distribution $P^{(m)}(y_1|y_\ominus; \varepsilon)$ depends effectively only on the components of $y_\ominus$ from nearby shells. When m lies sufficiently far inside the inertial interval, these shells also lie within the inertial interval. Statistical restoration of the hidden symmetry then implies that $P^{(m)}(y_1|y_\ominus; \varepsilon)$ is independent of m and coincides with the inertial-range conditional density $P(y_1|y_\ominus; \varepsilon)$ given by Eq. (107).

7.3 Perron–Frobenius modes

The universality of the conditional density established above implies that the recursive relation (118) defines the same linear transfer operator $\mathcal{L}_p(\varepsilon)$ for every reference shell sufficiently far inside the inertial interval,

$$\mathcal{L}_p(\varepsilon) : Q_p^{(m)}(\cdot; \varepsilon) \longmapsto Q_p^{(m+1)}(\cdot; \varepsilon). \tag{123}$$

The operator $\mathcal{L}_p(\varepsilon)$ is positive, in the sense that it maps positive measures to positive measures. We assume that this transfer operator possesses a simple dominant eigenvalue separated from

the remainder of the spectrum. Under this assumption, its asymptotic action is governed by the corresponding Perron–Frobenius eigenmode [23, 9],

$$Q_p^{(m)}(\cdot; \varepsilon) \sim \alpha_p(\varepsilon) \Lambda_p(\varepsilon)^m F_p(\cdot; \varepsilon), \quad m \rightarrow \infty, \quad (124)$$

where $\Lambda_p(\varepsilon) > 0$ is the Perron–Frobenius (dominant) eigenvalue, and $F_p(\cdot; \varepsilon)$ is the corresponding positive eigenfunction satisfying

$$\mathcal{L}_p(\varepsilon) F_p(\cdot; \varepsilon) = \Lambda_p(\varepsilon) F_p(\cdot; \varepsilon). \quad (125)$$

The coefficient $\alpha_p(\varepsilon)$ is not universal and depends only on the projection of the forcing-range statistics onto the dominant Perron–Frobenius eigenmode.

Substituting Eq. (124) into Eq. (115), and normalizing $F_p(\cdot; \varepsilon)$ to have unit mass, we obtain

$$S_p(m) \sim \alpha_p(\varepsilon) \lambda^{-mp/3} \Lambda_p(\varepsilon)^m. \quad (126)$$

Comparing Eq. (126) with the inertial-range scaling law $S_p(m) \propto k_m^{-\zeta_p}$, and using $k_m = \lambda^m$, we obtain

$$\zeta_p(\varepsilon) = \frac{p}{3} - \log_\lambda \Lambda_p(\varepsilon). \quad (127)$$

Thus, determining the anomalous scaling exponents reduces to computing the dominant Perron–Frobenius eigenvalues.

8 Perturbative Perron–Frobenius modes

We now use the perturbative multiplier statistics derived in Section 6 to solve the Perron–Frobenius eigenvalue problem (125) and determine the corresponding anomalous scaling exponents (127). Using the explicit form (118) of the transfer operator (123), the eigenvalue problem (125) becomes

$$(1 + \varepsilon y_1)^p P(y_1 | y_\ominus; \varepsilon) F_p(y_\ominus; \varepsilon) = \Lambda_p(\varepsilon) F_p(y'_\ominus; \varepsilon), \quad (128)$$

where $y'_\ominus$ and $(y_1, y_\ominus)$ are related by the shift $y'_i = y_{i+1}$, and $P(y_1 | y_\ominus; \varepsilon)$ is the inertial-range conditional density given by Eq. (107). We solve this eigenvalue problem order by order in ε , seeking the expansions

$$\Lambda_p(\varepsilon) = 1 + \varepsilon^2 \gamma_p^{(2)} + O(\varepsilon^4), \quad (129)$$

$$F_p(y_\ominus; \varepsilon) = P_0(y_\ominus) [1 + \varepsilon g_p^{(1)}(y_\ominus) + \varepsilon^2 g_p^{(2)}(y_\ominus) + O(\varepsilon^3)]. \quad (130)$$

The expansion of the eigenvalue in even powers of ε follows from the sign symmetry (87).

8.1 Zero- and first-order equations

At zeroth order, Eqs. (128)–(130) and (107) reduce to the shift-invariance relation (111), and are therefore identically satisfied. At first order, we obtain

$$g_p^{(1)}(y_\ominus) + py_1 + \sigma^{(1)}(y_1, y_\ominus) = g_p^{(1)}(y'_\ominus). \quad (131)$$

Substituting Eq. (108) and expressing $\rho^{(1)}(y_1, y_\ominus)$ using Eq. (110), we obtain

$$g_p^{(1)}(y_\ominus) - \rho^{(1)}(y_\ominus) + (p-1)y_1 = g_p^{(1)}(y'_\ominus) - \rho^{(1)}(y'_\ominus). \quad (132)$$

A formal solution of Eq. (132) is

$$g_p^{(1)}(y_\ominus) = \rho^{(1)}(y_\ominus) + (p-1) \sum_{i \leq 0} y_i. \quad (133)$$

Indeed, shifting the summation index gives $\sum_{i \leq 0} y'_i = y_1 + \sum_{i \leq 0} y_i$, which verifies Eq. (132). We note that only convergent local averages involving the expression (133) enter the eigenvalue calculation below.

8.2 Second-order equation

At second order, Eqs. (128)–(130), (107), and (111) yield

$$\begin{aligned} g_p^{(2)}(y_\ominus) + [py_1 + \sigma^{(1)}(y_1, y_\ominus)] g_p^{(1)}(y_\ominus) + \sigma^{(2)}(y_1, y_\ominus) \\ + py_1 \sigma^{(1)}(y_1, y_\ominus) + \frac{p(p-1)}{2} y_1^2 = g_p^{(2)}(y'_\ominus) + \gamma_p^{(2)}. \end{aligned} \quad (134)$$

Substituting Eqs. (108), (109), and (133) into Eq. (134), and collecting terms, we obtain

$$\begin{aligned} \gamma_p^{(2)} = g_p^{(2)}(y_\ominus) - g_p^{(2)}(y'_\ominus) + \rho^{(2)}(y_1, y_\ominus) - \rho^{(2)}(y_\ominus) + py_1 \rho^{(1)}(y_1, y_\ominus) \\ + (p-1) [py_1 + \rho^{(1)}(y_1, y_\ominus) - \rho^{(1)}(y_\ominus)] \sum_{i \leq 0} y_i + \frac{p(p-1)}{2} y_1^2. \end{aligned} \quad (135)$$

This equation determines the second-order eigenfunction correction $g_p^{(2)}$ and eigenvalue correction $\gamma_p^{(2)}$. The latter is obtained by taking the Gaussian average $\langle \cdot \rangle_0$ of Eq. (135). The averaged difference $g_p^{(2)}(y_\ominus) - g_p^{(2)}(y'_\ominus)$ vanishes by shift invariance of the Gaussian measure. The averaged difference $\rho^{(2)}(y_1, y_\ominus) - \rho^{(2)}(y_\ominus)$ also vanishes by marginalization over y_1 , according to Eq. (106). The same applies to $\rho^{(1)}(y_1, y_\ominus) - \rho^{(1)}(y_\ominus)$ in the term proportional to $\sum_{i \leq 0} y_i$, since this sum depends only on $y_\ominus$. We therefore obtain

$$\begin{aligned} \gamma_p^{(2)} = p \langle y_1 \rho^{(1)}(y_1, y_\ominus) \rangle_0 + p(p-1) \sum_{i \leq 0} \langle y_1 y_i \rangle_0 + \frac{p(p-1)}{2} \langle y_1^2 \rangle_0 \\ = pv_1 + p(p-1) \left(\sum_{j=1}^{\infty} \kappa_j + \frac{\kappa_0}{2} \right), \end{aligned} \quad (136)$$

where in the second equality we used Eqs. (105) and (94) to identify $\langle y_1 \rho^{(1)}(y_1, y_\Theta) \rangle_0 = v_1$, together with the covariance relation $\langle y_i y_j \rangle_0 = \kappa_{|i-j|}$ from Eqs. (89) and (91). Using v_1 from Eq. (72), the telescopic relation $\kappa_j = c_{j+1} - c_j$ from Eq. (92), and the conditions $c_0 = 0$ and $\lim_{i \rightarrow \infty} c_i = c_\infty$, we obtain

$$\begin{aligned} \gamma_p^{(2)} &= \frac{p}{3} (c_1 - \lambda^{1/3} - \lambda^{-1/3}) + p(p-1) \left(c_\infty - \frac{c_1}{2} \right) \\ &= -\frac{p(p-3)}{6} (c_1 - \lambda^{1/3} - \lambda^{-1/3}), \end{aligned} \quad (137)$$

where the second equality follows from Eq. (60). Substituting Eq. (137) into the eigenvalue expansion (129) and using Eq. (127), we obtain

$$\zeta_p(\varepsilon) = \frac{p}{3} + \frac{\varepsilon^2 p(p-3)}{6 \ln \lambda} (c_1 - \lambda^{1/3} - \lambda^{-1/3}) + O(\varepsilon^4). \quad (138)$$

Thus, Eq. (13) is recovered from the Perron–Frobenius formulation.

8.3 Comparison with numerical simulations

We now compare the analytical predictions with the results of direct numerical simulations. The anomalous scaling exponents are computed for $\varepsilon = 0.03, 0.06, \dots, 0.30$ by fitting the structure functions with power laws over the inertial interval of shells $n = 8, \dots, 18$; see Fig. 2(a).

The circles in Fig. 7 show the numerical anomalous corrections $\zeta_p - p/3$ as functions of ε^2 for the orders $p = 2, 3, 4, 6, 8$, and 10. The solid lines represent the leading-order theoretical prediction (138). As expected, the anomalous corrections depend linearly on ε^2 for sufficiently small ε , while visible deviations appear for larger values of ε , indicating the increasing importance of higher-order corrections.

To quantify this agreement, Fig. 8 compares the theoretical and numerical slopes of the anomalous corrections at $\varepsilon = 0$. The solid line shows the theoretical slope

$$\left. \frac{d\zeta_p}{d(\varepsilon^2)} \right|_{\varepsilon=0} = \frac{p(p-3)}{6 \ln \lambda} (c_1 - \lambda^{1/3} - \lambda^{-1/3}), \quad (139)$$

computed from Eq. (138). The circles denote the corresponding numerical slopes extracted from polynomial fits to the data in Fig. 7. The excellent agreement confirms the validity of the perturbative Perron–Frobenius analysis.

9 Conclusion

We have developed a first-principles perturbative derivation of anomalous scaling exponents for a nonlinear stochastic shell model of turbulence. The derivation combines the hidden-symmetry

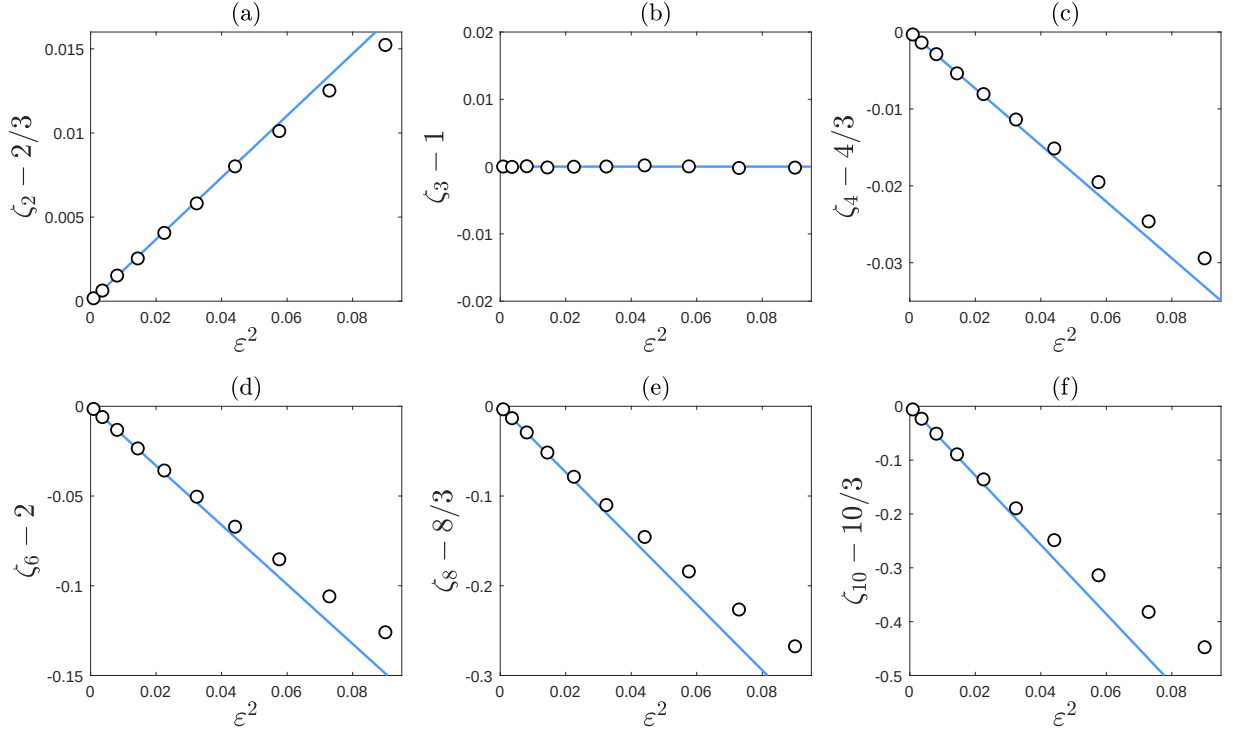


Figure 7: Numerically computed anomalous corrections $\zeta_p - p/3$ as functions of ε^2 for the orders (a) $p = 2$, (b) $p = 3$, (c) $p = 4$, (d) $p = 6$, (e) $p = 8$, and (f) $p = 10$ (circles). The solid lines show the leading-order perturbative prediction (138). The expected linear dependence on ε^2 is observed for sufficiently small ε , while deviations at larger values indicate higher-order corrections.

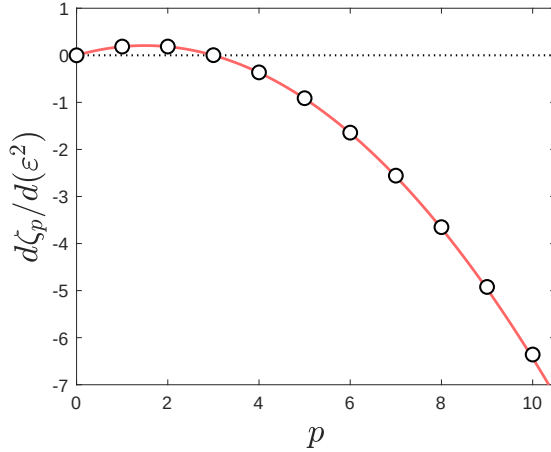


Figure 8: Derivative $d\zeta_p/d(\varepsilon^2)|_{\varepsilon=0}$ as a function of the structure-function order p . The solid line shows the leading-order perturbative prediction (139). The circles denote the numerical values extracted from polynomial fits to the data shown in Fig. 7.

formulation of the rescaled dynamics with the Perron–Frobenius formulation connecting structure functions to multiplier statistics. This framework yields an explicit analytical expression for the anomalous exponents in the weak-noise regime without invoking a closure of the correlation hierarchy. Direct numerical simulations independently verify the theoretical prediction.

The derivation is based on the assumption that the hidden symmetry is statistically restored in the inertial interval. This assumption replaces explicit infrared and ultraviolet boundary conditions by a statistical symmetry relating neighboring shells. The resulting perturbation theory is therefore formulated directly in the limit of infinite scale separation and yields the universal inertial-range statistics without introducing forcing- or dissipation-dependent cutoffs.

The framework developed here is not specific to the stochastic shell model considered in this work. The hidden symmetry underlying the present construction also arises in other shell models [27, 28, 30], in the Navier–Stokes equations [29, 33, 6], and in large-eddy simulation (LES) models [25]. Its statistical restoration has likewise been observed in turbulent regimes of these systems. This suggests that the present construction may provide a general analytical framework for deriving anomalous scaling exponents. A key challenge in extending the approach is to identify a suitable perturbative setting. In the present model, this is possible because the intermittent state emerges continuously from the deterministic K41 solution as the noise amplitude increases from zero.

An important open question is to understand why the hidden symmetry is statistically restored. One intriguing possibility is that this restoration is related to the renormalization-group structure of spontaneous stochasticity [32, 31]. In such a picture, the rescaled dynamics would converge toward a universal probability measure representing a statistical fixed point possessing the hidden symmetry, thereby providing a natural framework for studying its stability. Developing such a renormalization-group description remains an important direction for future work.

Appendix

A.1 Conversion to Itô form

For numerical integration, we use the Itô form of the stochastic shell model. Away from vanishing shell amplitudes, substituting Eq. (2) into Eq. (1) and applying the standard Stratonovich-to-Itô conversion yields the Itô equations below. For the first shell, we obtain

$$\begin{aligned} du_1 = & \left(1 - k_1 u_1 u_2 + \frac{1}{4} \varepsilon^2 k_1 \operatorname{sgn}(u_1) u_2^2 - \frac{1}{2} \varepsilon^2 k_1 |u_1| u_1 \right) dt \\ & - \varepsilon k_1^{1/2} |u_1|^{1/2} u_2 dw_1. \end{aligned} \tag{140}$$

For the interior shells $n = 2, \dots, N - 1$,

$$\begin{aligned} du_n = & \left(k_{n-1} u_{n-1}^2 - k_n u_n u_{n+1} - \frac{3}{4} \varepsilon^2 k_{n-1} |u_{n-1}| u_n \right. \\ & + \frac{1}{4} \varepsilon^2 k_n \operatorname{sgn}(u_n) u_{n+1}^2 - \frac{1}{2} \varepsilon^2 k_n |u_n| u_n \Big) dt \\ & + \varepsilon k_{n-1}^{1/2} |u_{n-1}|^{1/2} u_{n-1} dw_{n-1} - \varepsilon k_n^{1/2} |u_n|^{1/2} u_{n+1} dw_n. \end{aligned} \quad (141)$$

Finally, for the last shell,

$$\begin{aligned} du_N = & \left(k_{N-1} u_{N-1}^2 - D k_N |u_N| u_N - \frac{3}{4} \varepsilon^2 k_{N-1} |u_{N-1}| u_N \right) dt \\ & + \varepsilon k_{N-1}^{1/2} |u_{N-1}|^{1/2} u_{N-1} dw_{N-1}. \end{aligned} \quad (142)$$

Since the coefficient $|u_n|^{1/2}$ is not differentiable at $u_n = 0$, we adopt the resulting Itô equations, together with the convention $\operatorname{sgn}(0) = 0$, as the global definition of the stochastic dynamics.

A.2 Rescaled Stratonovich system

We first derive the equation for the rescaled shell variables. Using the ordinary Stratonovich chain rule together with Eq. (14), we obtain

$$\begin{aligned} dU_i = d \left(\frac{u_{m+i}}{|u_m|} \right) &= \frac{1}{|u_m|} \circ du_{m+i} - \frac{u_{m+i}}{|u_m|^2} \operatorname{sgn}(u_m) \circ du_m \\ &= \frac{u_{m+i-1}}{|u_m|} \circ dF_{m+i-1} - \frac{u_{m+i+1}}{|u_m|} \circ dF_{m+i} \\ &\quad - \frac{u_{m+i}}{|u_m|} \operatorname{sgn}(u_m) \left(\frac{u_{m-1}}{|u_m|} \circ dF_{m-1} - \frac{u_{m+1}}{|u_m|} \circ dF_m \right). \end{aligned} \quad (143)$$

This yields Eq. (19) after using Eqs. (17), (18) and $dG_i = dF_{m+i}$. It remains to express the transfer processes G_i in terms of the rescaled time τ .

We next recall the general transformation rule for a state-dependent time change. Let the state X satisfy

$$dX = b(X) dt + \sum_n \sigma_n(X) \circ dw_n, \quad (144)$$

and consider another process driven by the same Wiener processes,

$$dY = c(X) dt + \sum_n \eta_n(X) \circ dw_n. \quad (145)$$

Under the time change

$$d\tau = a(X) dt, \quad d\tilde{w}_n = a(X)^{1/2} dw_n, \quad a(X) > 0, \quad (146)$$

the Stratonovich equation for Y becomes

$$dY = \left[\frac{c(X)}{a(X)} + \frac{1}{4a(X)^2} \sum_n \eta_n(X) (\sigma_n(X) \cdot \nabla a(X)) \right] d\tau + \sum_n \frac{\eta_n(X)}{a(X)^{1/2}} \circ d\tilde{w}_n. \quad (147)$$

One can verify these relations by converting the equations to Itô form, performing the time change, and converting the resulting equations back to Stratonovich form.

We apply this rule with $X = (u_n)$ and $a(u) = k_m|u_m|$. The second process is $Y = F_{m+i}$, which satisfies

$$dF_{m+i} = k_{m+i}u_{m+i} dt + \varepsilon k_{m+i}^{1/2}|u_{m+i}|^{1/2} \circ dw_{m+i}. \quad (148)$$

Since the clock depends only on u_m ,

$$\sigma_n(u) \cdot \nabla a(u) = k_m \operatorname{sgn}(u_m) \sigma_{n,m}(u), \quad (149)$$

where $\sigma_{n,m}$ denotes the coefficient of dw_n in the equation for du_m ; see Eqs. (14) and (2). Only the noises w_{m-1} and w_m contribute, with

$$\sigma_{m-1,m} = \varepsilon k_{m-1}^{1/2}|u_{m-1}|^{1/2}u_{m-1}, \quad \sigma_{m,m} = -\varepsilon k_m^{1/2}|u_m|^{1/2}u_{m+1}. \quad (150)$$

From Eq. (148), the only nonzero coefficient is

$$\eta_{m+i} = \varepsilon k_{m+i}^{1/2}|u_{m+i}|^{1/2}. \quad (151)$$

Moreover, $d\tilde{w}_{m+i} = (k_m|u_m|)^{1/2}dw_{m+i} = dW_i$ by Eq. (21). Applying Eq. (147), we write $dG_i = dF_{m+i}$ in the rescaled time as

$$\begin{aligned} dG_i = & \left[\lambda^i \frac{u_{m+i}}{|u_m|} + \frac{\varepsilon^2}{4} \operatorname{sgn}(u_m) \left(\lambda^{-1} \frac{|u_{m-1}|u_{m-1}}{|u_m|^2} \delta_{i,-1} - \frac{u_{m+1}}{|u_m|} \delta_{i0} \right) \right] d\tau \\ & + \varepsilon \lambda^{i/2} \frac{|u_{m+i}|^{1/2}}{|u_m|^{1/2}} \circ dW_i. \end{aligned} \quad (152)$$

Using Eqs. (17) and (18), this becomes Eq. (20).

We finally verify the hidden-symmetry transformation (24) directly from the definition (17) of the rescaled variables. Changing the reference shell from m to $m+1$, the rescaled shell variables become

$$U'_i = \frac{u_{m+1+i}}{|u_{m+1}|} = \frac{u_{m+i+1}/|u_m|}{|u_{m+1}|/|u_m|} = \frac{U_{i+1}}{|U_1|}. \quad (153)$$

Similarly, the rescaled time associated with the new reference shell satisfies

$$d\tau' = k_{m+1}|u_{m+1}| dt = \lambda|U_1| k_m|u_m| dt = \lambda|U_1| d\tau. \quad (154)$$

Finally, using the definition (21) of the rescaled Wiener processes, we obtain

$$dW'_i = (k_{m+1}|u_{m+1}|)^{1/2} dw_{m+1+i} = (\lambda|U_1|)^{1/2} (k_m|u_m|)^{1/2} dw_{m+i+1} = (\lambda|U_1|)^{1/2} dW_{i+1}. \quad (155)$$

Since the rescaled equations (19) and (20) have the same form for an arbitrary choice of the reference shell, changing m to $m+1$ maps a solution of the rescaled dynamics to a solution of the same equations. This establishes the hidden symmetry (24).

A.3 Expansion of the rescaled Itô system

For the perturbative analysis, we require the Itô form of the rescaled dynamics restricted to the positive branch $U_i > 0$ with $U_0 \equiv 1$. Applying the standard Stratonovich–Itô conversion to Eqs. (19) and (20), we obtain Eq. (28) with

$$\begin{aligned} a_i(Z; \varepsilon) = & \varepsilon^{-1} \lambda^{i/3} \left[\lambda^{i-1} U_{i-1}^2 - \lambda^i U_i U_{i+1} - U_i (\lambda^{-1} U_{-1}^2 - U_1) \right. \\ & + \varepsilon^2 \left(-\frac{3}{4} \lambda^{i-1} U_{i-1} U_i + \frac{1}{4} \lambda^i U_{i+1}^2 - \frac{1}{2} \lambda^i U_i^2 + \frac{3}{4} \lambda^{-1} U_i U_{-1} \right. \\ & \left. \left. + \frac{1}{2} U_i + \lambda^{-1} U_i U_{-1}^3 + \frac{3}{4} U_i U_1^2 + \lambda^{-1} U_{-1}^2 \delta_{i,-1} + U_1 \delta_{i1} \right) \right], \end{aligned} \quad (156)$$

where $U_i = \lambda^{-i/3} (1 + \varepsilon Z_i)$, while the noise coefficients are

$$\begin{aligned} B_{ij}(Z; \varepsilon) = & \lambda^{i/3} \left[\lambda^{(i-1)/2} U_{i-1}^{3/2} \delta_{j,i-1} - \lambda^{i/2} U_i^{1/2} U_{i+1} \delta_{j,i} \right. \\ & \left. - U_i (\lambda^{-1/2} U_{-1}^{3/2} \delta_{j,-1} - U_1 \delta_{j0}) \right]. \end{aligned} \quad (157)$$

Expanding Eqs. (156) and (157) in powers of ε yields Eqs. (29) with

$$(A^{(0)} Z)_i = \lambda^{(2i-1)/3} (2Z_{i-1} - Z_i - Z_{i+1}) - \lambda^{-1/3} (2Z_{-1} - Z_1), \quad (158)$$

$$\begin{aligned} a_i^{(1)}(Z) = & \lambda^{(2i-1)/3} (Z_{i-1}^2 - Z_i Z_{i+1}) - \lambda^{-1/3} (Z_{-1}^2 + 2Z_i Z_{-1} - Z_i Z_1) \\ & - \frac{1}{2} \lambda^{2(i-1)/3} - \frac{1}{2} \lambda^{2i/3} + \frac{3}{2} + \frac{3}{2} \lambda^{-2/3} \\ & + \delta_{i,1} + \lambda^{-2/3} \delta_{i,-1} - (1 + \lambda^{-2/3}) \delta_{i,0}, \end{aligned} \quad (159)$$

$$B_{ij}^{(0)} = \lambda^{i/3} \delta_{j,i-1} - \lambda^{(i-1)/3} \delta_{j,i} - \delta_{j,-1} + \lambda^{-1/3} \delta_{j,0}, \quad (160)$$

$$\begin{aligned} B_{ij}^{(1)}(Z) = & \frac{3}{2} \lambda^{i/3} Z_{i-1} \delta_{j,i-1} - \lambda^{(i-1)/3} \left(Z_{i+1} + \frac{1}{2} Z_i \right) \delta_{j,i} \\ & - \left(Z_i + \frac{3}{2} Z_{-1} \right) \delta_{j,-1} + \lambda^{-1/3} (Z_i + Z_1) \delta_{j,0}. \end{aligned} \quad (161)$$

These formulas satisfy the normalization $Z_0 \equiv 0$, since

$$(A^{(0)} Z)_0 = a_0^{(1)}(Z) = 0, \quad B_{0j}^{(0)} = B_{0j}^{(1)}(Z) = 0. \quad (162)$$

The diffusion matrix admits the expansion (30), with

$$D^{(0)} = B^{(0)} (B^{(0)})^T, \quad D^{(1)}(Z) = B^{(0)} (B^{(1)}(Z))^T + B^{(1)}(Z) (B^{(0)})^T. \quad (163)$$

Thus, $D^{(0)}$ is constant, whereas $D^{(1)}(Z)$ is linear in Z .

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